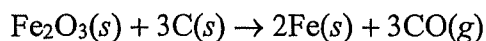


Physical constants

Bohr radius:	$a_0 = 5.29 \times 10^{-11} \text{ m}$;	Boltzmann's constant:	$k_B = 1.38 \times 10^{-23} \text{ J K}^{-1}$
Electron charge:	$e = 1.60 \times 10^{-19} \text{ C}$;	Faraday constant:	$F = 9.65 \times 10^4 \text{ C mol}^{-1}$
Planck's constant:	$h = 6.63 \times 10^{-34} \text{ J s}$;	Speed of light in vacuum:	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
Mass of electron:	$m_e = 9.11 \times 10^{-31} \text{ kg}$;	Mass of proton:	$m_p = 1.67 \times 10^{-27} \text{ kg}$
Mass of neutron:	$m_n = 1.67 \times 10^{-27} \text{ kg}$		
Universal gas constant: $R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1} = 0.082 \text{ L atm mol}^{-1} \text{ K}^{-1}$			
$\ln 2 = 0.6931$; $\ln 3 = 1.0986$; $\ln 7 = 1.9459$; $\pi = 3.14$			

一、單選題 (每題 2.5 分, 共計 100 分), 答錯不倒扣

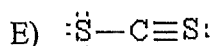
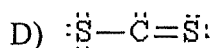
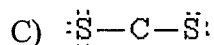
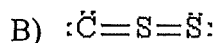
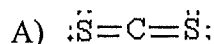
1. Iron oxide can be reduced to iron by a reaction with carbon to form carbon monoxide



If 95.0 grams of iron oxide is reacted with excess carbon yields 63.0 g of iron, what is the percent yield of this reaction?

- A) 12% B) 60% C) 75% D) 85% E) 95%

2. Choose the electron dot formula that most accurately describes the bonding in CS_2 .



3. An electron in a 10.0-nm one-dimensional box is excited from the ground state into a higher energy state by absorbing a photon with wavelength $1.374 \times 10^{-5} \text{ m}$. Determine the final energy level for this transition.

- A) $n = 2$ B) $n = 3$ C) $n = 4$ D) $n = 5$ E) $n = 6$

4. For H_2 , which of the following statements is not true?

- A) Its HOMO is the σ_{1s} orbital.
 B) Its LUMO is the σ_{1s}^* orbital.
 C) Its lowest-energy excited state electron configuration is $(\sigma_{1s})^1(\sigma_{1s}^*)^1$.
 D) The bond order for Its lowest-energy excited state is 1.
 E) Its ground state electron configuration is $(\sigma_{1s})^2$.

5. Which of the following statements is(are) true?

- I. An excited atom can return to its ground state by absorbing electromagnetic radiation.
- II. The energy of an atom is increased when electromagnetic radiation is emitted from it.
- III. The energy of electromagnetic radiation increases as its frequency increases.
- IV. An electron in the $n = 4$ state in the hydrogen atom can go to the $n = 2$ state by emitting electromagnetic radiation at the appropriate frequency.
- V. The frequency and wavelength of electromagnetic radiation are inversely proportional to each other.

A) II, III, IV B) III, V C) I, II, III D) I, II, IV E) III, IV, V

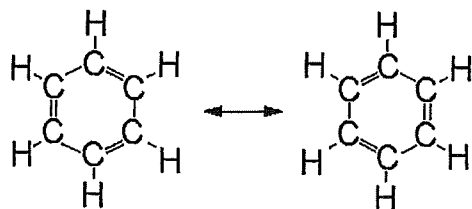
6. Rank from smallest to largest in terms of atom/ionic radii.

- A) S^{2-} , Cl^- , Ar, K^+ , Ca^{2+}
- B) Ca^{2+} , K^+ , Ar, Cl^- , S^{2-}
- C) Ca^{2+} , S^{2-} , K^+ , Cl^- , Ar
- D) Ar, K^+ , Cl^- , Ca^{2+} , S^{2-}
- E) Ar, Cl^- , K^+ , S^{2-} , Ca^{2+}

7. Which of the following statements is true concerning ideal gases?

- A) The temperature of the gas sample is directly related to the average velocity of the gas particles.
- B) At STP, 1.0 L of Ar(g) contains about twice the number of atoms as 1.0 L of Ne(g) because the molar mass of Ar is about twice that of Ne.
- C) A gas exerts pressure as a result of the collisions of the gas molecules with the walls of the container.
- D) The gas particles in a sample exert attraction on one another.
- E) All of these statements are false.

8. A Lewis dot structure for benzene, C_6H_6 would give two equivalent resonance structures for this cyclic molecule with alternating single and double bonds



A combined LCAO and MO picture for the bonding of the carbon atoms in benzene would be

- A) $1\sigma(Csp^2-Csp^2)$ bonds between each C atom, and 3π MO delocalized over the C's
- B) $2\sigma(Csp^3-Csp^3)$ bonds between each C atom, and 1π MO delocalized over the C's
- C) $1\sigma(Csp^3-Csp^3)$ bonds between each C atom, and 1π MO delocalized over the C's
- D) $1\sigma(Csp^2-Csp^2)$ bonds between each C atom, and 2π MO delocalized over the C's
- E) $3\sigma(C2p-C2p)$ bonds between each C atom

注意：背面有試題

9. Thyroxine, an important hormone that controls the rate of metabolism in the body, can be isolated from the thyroid gland. If 0.455 g of thyroxine is dissolved in 10.0 g of benzene, the freezing point of the solution is 5.144°C . Pure benzene freezes at 5.444°C and has a value for the molal freezing-point-depression constant of K_f of $5.12^{\circ}\text{C}/m$. What is the molar mass of thyroxine?

- A) 7770 g/mol
- B) 777 g/mol
- C) 2330 g/mol
- D) 285 g/mol
- E) 3760 g/mol

10. Which of the following shows these molecules in order from most polar to least polar?

- A) $\text{CH}_4 > \text{CF}_2\text{Cl}_2 > \text{CF}_2\text{H}_2 > \text{CCl}_4 > \text{CCl}_2\text{H}_2$
- B) $\text{CH}_4 > \text{CF}_2\text{H}_2 > \text{CF}_2\text{Cl}_2 > \text{CCl}_4 > \text{CCl}_2\text{H}_2$
- C) $\text{CF}_2\text{H}_2 > \text{CCl}_2\text{H}_2 > \text{CF}_2\text{Cl}_2 > \text{CH}_4 = \text{CCl}_4$
- D) $\text{CF}_2\text{Cl}_2 > \text{CF}_2\text{H}_2 > \text{CCl}_2\text{H}_2 > \text{CH}_4 = \text{CCl}_4$
- E) $\text{CF}_2\text{Cl}_2 > \text{CF}_2\text{H}_2 > \text{CCl}_4 > \text{CCl}_2\text{H}_2 > \text{CH}_4$

11. Which of the following statements are true about starch?

- I. The monomers are fructose and glucose.
 - II. The monomer is glucose.
 - III. It is the main carbohydrate reservoir in plants.
 - IV. It is the main carbohydrate reservoir in animals.
 - V. It is an addition polymer.
 - VI. It is a condensation polymer.
- A) I, III, IV, V B) II, IV, VI C) II, III, VI D) I, III, V E) I, IV, V

12. The octahedral compound $[\text{FeCl}_3(\text{NH}_3)_3]$ is found to be paramagnetic. This means

- A) it must be a high spin complex.
- B) it must be a low spin complex.
- C) it could be either a high spin or low spin complex.
- D) it has a crystal field stabilization that is zero.
- E) a & d

13. Which one of the following statements about the structure of proteins is *incorrect*?

- A) Disulfide bonds provide strong intrachain interactions.
- B) Hydrogen bonding stabilizes the α -helix proteins.
- C) Nonpolar groups tend to face the outside of a protein in an aqueous solution.
- D) Ionized amino acid side chains can form salt bridges within a protein.
- E) Heat can disrupt tertiary structure.

14. Helium effuses through a small opening at a rate of $1 \times 10^{-9} \text{ mol s}^{-1}$. An unknown gas at the same temperature and pressure is found to effuse through the same opening at a rate of $3.78 \times 10^{-10} \text{ mol s}^{-1}$. What is the molecular mass of the unknown gas?
A) 4 g mol^{-1} B) 14 g mol^{-1} C) 18 g mol^{-1} D) 28 g mol^{-1} E) 40 g mol^{-1}
15. It is desired to determine the concentration of arsenic in a lake sediment sample by means of neutron activation analysis. The nuclide $^{75}_{33}\text{As}$ captures a neutron to form $^{76}_{33}\text{As}$, which in turn undergoes β decay. The daughter nuclide produces the characteristic γ rays used for the analysis. What is the daughter nuclide?
A) $^{75}_{34}\text{Se}$ B) $^{76}_{32}\text{Ge}$ C) $^{74}_{31}\text{Ga}$ D) $^{76}_{34}\text{Se}$ E) $^{74}_{34}\text{Se}$
16. What is the complementary nucleic acid sequence for the DNA sequence
GAC TAC GTT GGC?
A) GAC TAC GTT GGC
B) TCA GCA TGG CTA
C) CGA GTG CAT CAG
D) GCG AAG GGG TTG
E) CTG ATG CAA CCG
17. The complex ions of Zn^{2+} are all colorless. The most likely explanation for this is that
A) Zn^{2+} is paramagnetic.
B) Zn^{2+} exhibits "d orbital" splittings in its complexes such that they absorb all wavelengths in the visible region.
C) because Zn^{2+} is a d^{10} ion, it does not absorb visible light even though the "d orbital" splittings are correct for absorbing visible wavelengths.
D) Zn^{2+} is not a transition metal ion.
E) because Zn^{2+} is a d^8 ion, it does not absorb visible light even though the "d orbital" splitting are correct for absorbing visible wavelengths.
18. Acetone (mw = 58.08, $P^*_{25^\circ\text{C}} = 232 \text{ mmHg}$) and butanone (mw = 72.11, $P^*_{25^\circ\text{C}} = 100 \text{ mmHg}$) have the indicated molar masses and vapor pressures. A container holds 1.00 kg of butanone. How much acetone must be added to the butanone to elevate the total vapor pressure over the mixture to 125 mmHg at 25°C ?
A) 88 kg B) 188 g C) 218 g D) 288g E) 388g

19. In a certain reversible expansion, a system at 300. K absorbs exactly 6.00×10^2 J of heat. In the irreversible recompression to the original state of the system, twice as much work is done on the system as is performed on the surroundings in the expansion. What is the entropy change of the system in the recompression step?

- A) -4.00 J/K B) -2.00 J/K C) 0.00 J/K D) 2.00 J/K E) 4.00 J/K

20. A certain compound with a molar mass of 138.0 g/mol crystallizes with the sodium chloride (rock salt) structure. The length of an edge of the unit cell is 488 pm. What is the density of this compound?

- A) 7.89 g/cm³
B) 1.97 g/cm³
C) 4.75 g/cm³
D) 1.19 g/cm³
E) 1.88 g/cm³

21. Name the following compound, $\text{Pt}(\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2)\text{Cl}_4\text{Cl}_2$

- A) hexachloro(ethylenediamine)platinum (IV)
B) hexachloride(ethylenediamine)platinum (VI)
C) tetrachloro(ethylenediamine)platinum (IV) chloride
D) tetrachloro(ethylenediamine)platinum (VI) chloride
E) Platinum(ethylenediamine)hexachloride

22. The following data were collected for the decay of HO_2 radicals.

Time	$[\text{HO}_2]$	Time	$[\text{HO}_2]$
0 s	1.0×10^{11} molec/cm ³	14 s	1.25×10^{10} molec/cm ³
2 s	5.0×10^{10} molec/cm ³	30 s	6.225×10^9 molec/cm ³
6 s	2.5×10^{10} molec/cm ³		

Which of the following statements is true?

- A) The decay of HO_2 occurs by a first-order process.
B) The half-life of the reaction is 2 ms.
C) A plot of $\ln [\text{HO}_2]$ versus time is linear with a slope of $-k$.
D) A plot of $1/[\text{HO}_2]$ versus time gives a straight line.
E) The rate of the reaction increases with time.

23. The energy expressions for the electrons in the He^+ ion and the hydrogen atom are $E_n(\text{H}) = -a/n^2$ and $E_n(\text{He}^+) = -4a/n^2$

Which of the following statements is(are) correct?

- I. For the transitions $n_1 \rightarrow n_2$, the frequency is larger for H than for He^+ .
- II. The first ionization energy of the H atom is smaller than the second ionization energy of the He atom.
- III. The 1s orbital in He^+ is larger (in the sense that the probability density is shifted outward) than the 1s orbital in H.

A) I only B) II only C) III only D) I and II only E) I, II, and III

24. Consider the following gas samples:

Sample A	Sample B
$\text{S}_2(\text{g})$	$\text{O}_2(\text{g})$
$n = 1 \text{ mol}$	$n = 2 \text{ mol}$
$T = 800 \text{ K}$	$T = 400 \text{ K}$
$P = 0.20 \text{ atm}$	$P = 0.40 \text{ atm}$

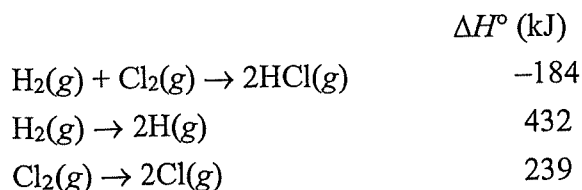
Which one of the following statements is *false*?

- A) The volume of sample A is twice the volume of sample B.
- B) The average kinetic energy of the molecules in sample A is twice the average kinetic energy of the molecules in sample B.
- C) The root-mean-square velocity of molecules in sample A is twice as large as the root-mean-square velocity of molecules in sample B.
- D) The fraction of molecules in sample A having a kinetic energies greater than some high fixed value is larger than the fraction of molecules in sample B having kinetic energies greater than that same high fixed value.
- E) Assuming identical intermolecular forces in the two samples, sample A should be more nearly ideal than sample B.

25. What reason is given for the stability of C—C, N—N, and O—O bonds, compared to the instability of Si—Si, P—P, and S—S bonds?

- A) Their metallic character varies greatly.
- B) There are large differences in their ionization energies.
- C) There are large differences in their abilities to form strong pi bonds.
- D) There are large differences in their electronegativities.
- E) none of these

26. Using the following data reactions:



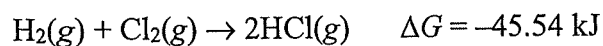
calculate the energy of an H-Cl bond.

- A) 770 kJ B) 856 kJ C) 518 kJ D) 326 kJ E) 428 kJ

27. Calculate $[\text{H}^+]$ in a 0.40 M solution of NaH_2PO_4 . (For H_3PO_4 , $K_{a1} = 7.5 \times 10^{-3}$, $K_{a2} = 6.2 \times 10^{-8}$, $K_{a3} = 4.8 \times 10^{-13}$)

- A) $2.2 \times 10^{-5} \text{ M}$ B) $1.6 \times 10^{-4} \text{ M}$ C) 0.40 M D) $2.5 \times 10^{-4} \text{ M}$ E) $3.9 \times 10^{-11} \text{ M}$

28. A mixture of hydrogen and chlorine remains unreacted until it is exposed to ultraviolet light from a burning magnesium strip. Then the following reaction occurs very rapidly.



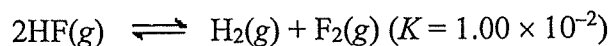
$$\Delta H = -44.12 \text{ kJ}$$

$$\Delta S = -4.76 \text{ J/K}$$

Select the statement below that best explains this behavior.

- A) The reactants are thermodynamically more stable than the products.
 B) The reaction has a small equilibrium constant.
 C) The ultraviolet light raises the temperature of the system and makes the reaction more favorable.
 D) The negative value for ΔS slows down the reaction.
 E) The reaction is spontaneous, but the reactants are kinetically stable.

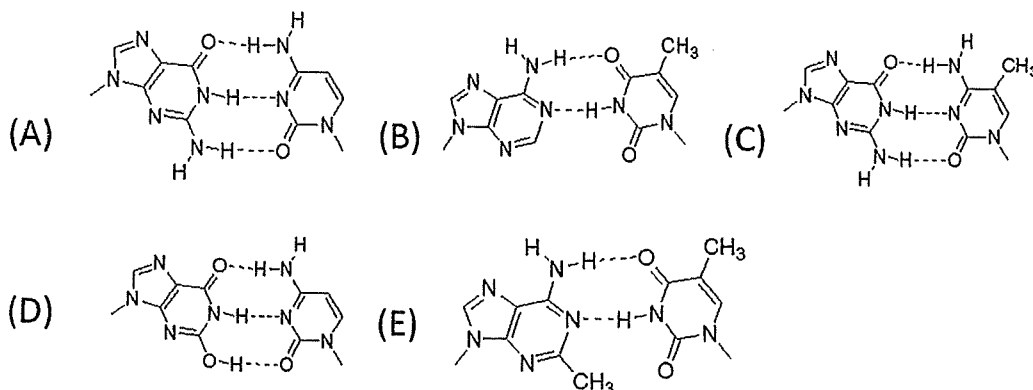
29. Consider the following reaction:



Given 1.22 mol of $\text{HF}(\text{g})$, 0.760 mol of $\text{H}_2(\text{g})$, and 1.09 mol of $\text{F}_2(\text{g})$ are mixed in a 4.00-L flask, determine the reaction quotient, Q , and the net direction to achieve equilibrium.

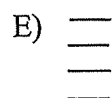
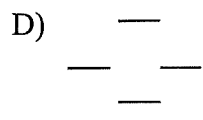
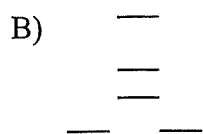
- A) $Q = 0.679$; the equilibrium shifts to the right.
 B) $Q = 0.557$; the equilibrium shifts to the left.
 C) $Q = 0.679$; the equilibrium shifts to the left.
 D) $Q = 0.557$; the equilibrium shifts to the right.
 E) $Q = 1.43$; the system is at equilibrium.

30. Which statement is true of a process in which 1 mol of a gas is expanded from state A to state B?
- A) When the gas expands from state A to state B, the surroundings are doing work on the system.
 B) The amount of work done in the process must be the same, regardless of the path.
 C) It is not possible to have more than one path for a change of state.
 D) The final volume of the gas will depend on the path taken.
 E) The amount of heat released in the process will depend on the path taken.
31. An ethyl group (CH_3CH_2^-) that is attached to a substituent that does not contain a hydrogen atom appears as what in a NMR spectrum?
- A) a triplet and a quartet with relative intensities of 3 and 2, respectively
 B) a doublet and a triplet with relative intensities of 3 and 2, respectively
 C) a triplet and a quartet with relative intensities of 2 and 3, respectively
 D) a doublet and a triplet with relative intensities of 2 and 3, respectively
 E) none of these
32. Which base-pair is most stable in Watson-Crick DNA Structure?

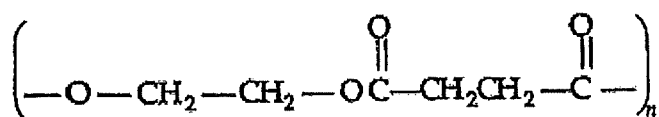


33. A 0.4987 g sample of a compound known to contain only carbon, hydrogen, and oxygen was burned in oxygen to yield 0.9267 g of CO_2 and 0.1897 g of H_2O . What is the empirical formula of the compound?
- A) CHO B) $\text{C}_3\text{H}_3\text{O}_2$ C) $\text{C}_2\text{H}_2\text{O}$ D) $\text{C}_6\text{H}_3\text{O}_2$ E) $\text{C}_3\text{H}_6\text{O}_2$
34. When a student performs an endothermic reaction in a calorimeter, how (if any) does the calculated value of ΔH differ from the actual value if the heat exchanged with the calorimeter is not taken into account?
- A) ΔH_{calc} is less positive because the reaction absorbs heat from the calorimeter.
 B) ΔH_{calc} is less negative because the calorimeter absorbs heat from the reaction.
 C) ΔH_{calc} is more positive because the reaction absorbs heat from the calorimeter.
 D) ΔH_{calc} is more negative because the calorimeter always absorbs heat from the reaction.
 E) ΔH_{calc} equals the actual value because the calorimeter does not absorb heat.

35. Choose the most likely pattern for the crystal field diagram for the complex *trans*- $[\text{Ni}(\text{NH}_3)_2(\text{CN})_4]^{2-}$ where CN^- produces a much stronger crystal field than does NH_3 .



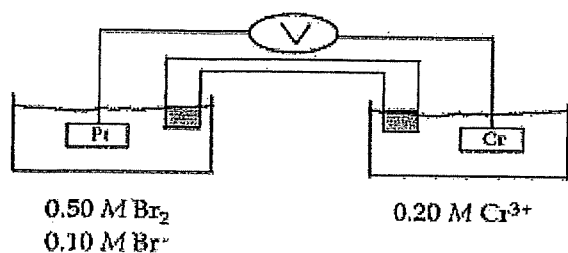
36. What monomer(s) is(are) needed to make the polymer shown here?



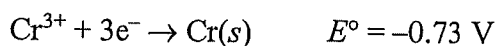
- I. $\text{HOCH}_2\text{CH}_2\text{OH}$
- II. $\text{HOOCCH}_2\text{CH}_2\text{COOH}$
- III. $\text{HOCH}_2\text{CH}_2\text{COOH}$
- IV. $\text{HOCH}=\text{CHOH}$
- V. $\text{HOOCCH}=\text{CHCOOH}$

A) II only B) III only C) I and II D) IV and V E) II and III

37. Consider the galvanic cell shown below (the contents of each half-cell are written beneath each compartment).



The standard reduction potentials are as follows:



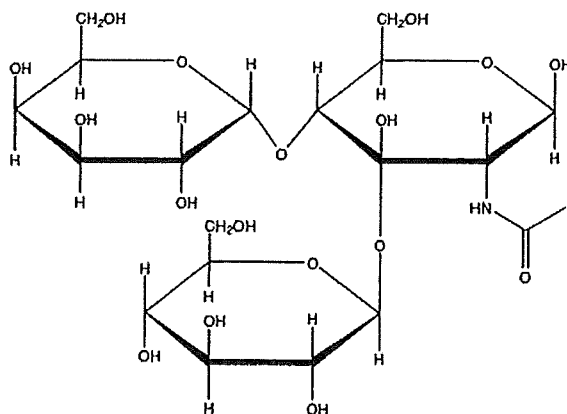
2Br^-

What is the value of E for this cell at 25°C ?

- A) 2.21 V B) 1.76 V C) 2.12 V D) 1.88 V E) 0.59 V

38. What glycosidic bonds are seen in the following portion of an *N*-linked oligosaccharide?

Abbreviations: galactose: Gal; glucose: Glu; *N*-acetylglucosamine: NAG



- A) $\alpha(1 \rightarrow 4)$ between Gal and NAG; $\beta(1 \rightarrow 6)$ between Glu and NAG
 B) $\beta(1 \rightarrow 4)$ between Gal and NAG; $\alpha(1 \rightarrow 3)$ between Glu and NAG
 C) $\beta(1 \rightarrow 3)$ between Gal and NAG; $\alpha(1 \rightarrow 4)$ between Glu and NAG
 D) $\beta(1 \rightarrow 4)$ between Gal and NAG; $\beta(1 \rightarrow 3)$ between Glu and NAG
 E) $\alpha(1 \rightarrow 4)$ between Gal and NAG; $\beta(1 \rightarrow 3)$ between Glu and NAG

39. The pK_a of a certain weak acid is 4.0. Calculate the ratio of proton acceptor to proton donor at $\text{pH } 7.0$.

- A) 1000:1 B) 100:1 C) 20:1 D) 3:1 E) 1:1

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40. Which group shows the correct order of first ionization energy?

- A) $\text{Na} > \text{P} > \text{Cl}$
- B) $\text{Cs} > \text{Na} > \text{K}$
- C) $\text{K} > \text{Ca} > \text{Ge}$
- D) $\text{Cs} < \text{Rb} < \text{Na}$
- E) $\text{Al} > \text{Si} > \text{P}$

注意：背面有試題